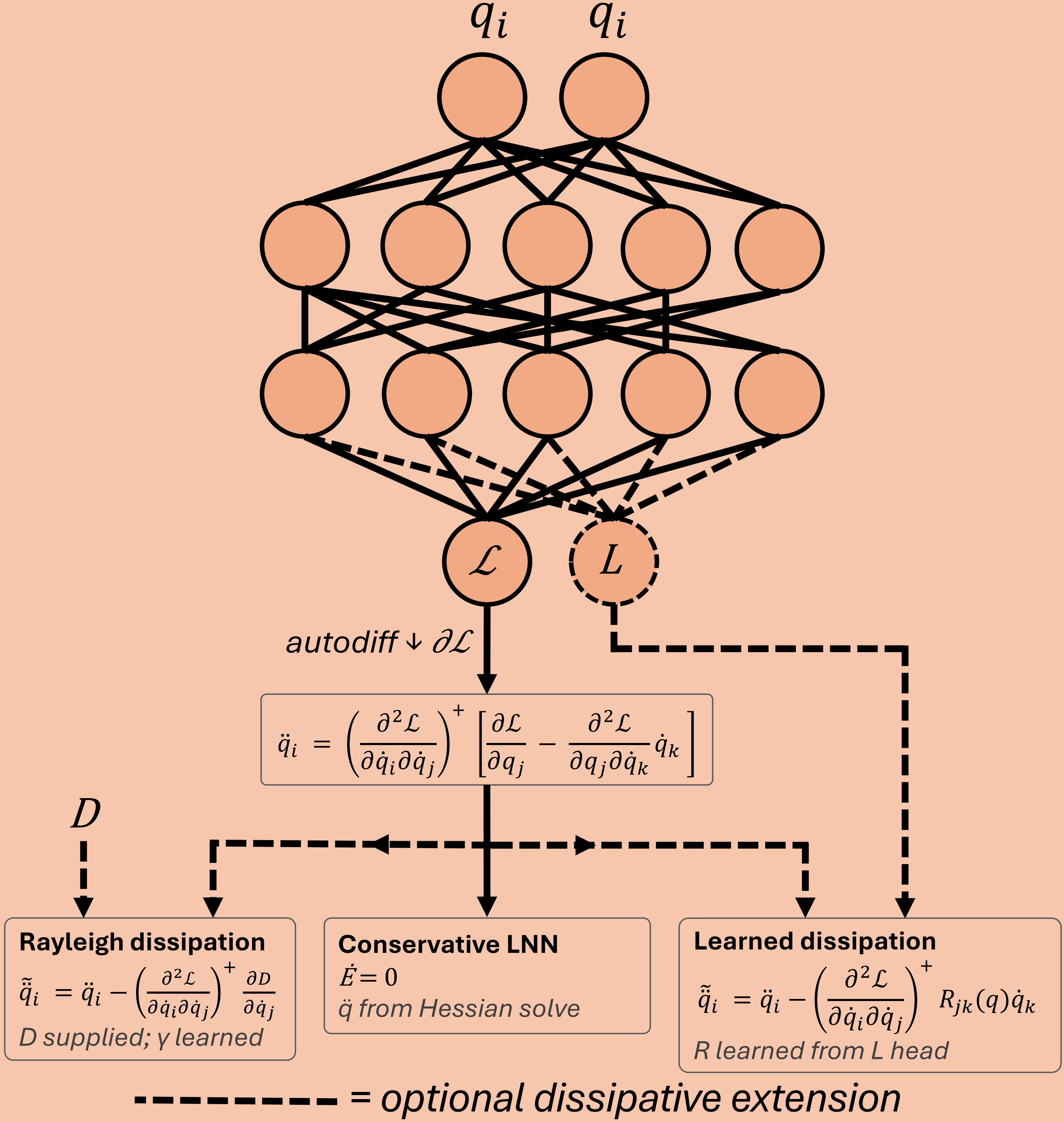

q_i
$\dot{q}_i$
$\mathcal{L}$
L
autodiff ↓ $\partial\mathcal{L}$
$\ddot{q}_i = \left(\frac{\partial^2\mathcal{L}}{\partial\dot{q}_i\partial\dot{q}_j}\right)^+ \left[\frac{\partial\mathcal{L}}{\partial q_j} - \frac{\partial^2\mathcal{L}}{\partial q_j\partial\dot{q}_k}\dot{q}_k\right]$
D
Rayleigh dissipation
$\tilde{\ddot{q}}_i = \ddot{q}_i - \left(\frac{\partial^2\mathcal{L}}{\partial\dot{q}_i\partial\dot{q}_j}\right)^+ \frac{\partial D}{\partial\dot{q}_j}$
D supplied; γ learned
Conservative LNN
$\dot{E} = 0$
q̈ from Hessian solve
Learned dissipation
$\tilde{\ddot{q}}_i = \ddot{q}_i - \left(\frac{\partial^2\mathcal{L}}{\partial\dot{q}_i\partial\dot{q}_j}\right)^+ R_{jk}(q)\dot{q}_k$
R learned from L head
= optional dissipative extension